Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

Bachelor in Computer Applications

Course Title: **Artificial Intelligence**

Code No: Semester: **4**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

***Note:** Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.*

Group B

1. What is knowledge, representation and reasoning? When a machine is said to be passed Turing test? Give any two examples of constraint satisfaction problem.

2. Why CNF is necessary? From the following facts, show that "Charlie is a mammal" using FOPL based resolution method.• Cows, pigs and horses are mammals• The child of a horse is a horse• Bluebeard is a horse• Bluebeard is Charlie's father• Child and father are inverse relations• Every mammal has a father.

3. Justify that AI can't exist without searching? Solve the following 8 puzzle problem using Hill climbing search algorithm.

4. Explain the different ambiguities related to NLP.

5. Differentiate between goal based agent and utility based agent.

6. Describe the concept of learning by analogy with an example.

7. What is the task of activation function? What are its types?

8. How Dempster - Shafer theory can be used for reasoning with certainty? Explain with an example.

9. Define expert system. Describe its architecture.

10. What is the purpose of alpha beta pruning? Explain.

11. Illustrate your own scenario for shopping at my mall and represent those concepts using scripts.

12. Define deterministic and non-deterministic environment. Differentiate between BFS and DFS.